Solution :-

1. A wave is not motion of matter as a whole in a medium.

2. X - rays are EMW so doesn't required medium.

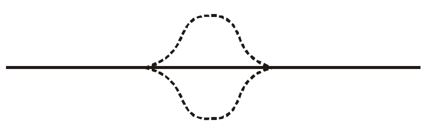
4. As the wave passes through gas it compress or expands a small region of gas. This cause change in density of that region this change induce a change in pressure.

5. Spring is a solid material & in solid we can generate both type of waves.

6. Solid & Liquid are much more difficult to compress than gases and so have much higher value of Bulk Modulus. This factor compensates for their higher densities than gases.

7. When wave reflects from denser medium or rigid boundary there is a phase change of π on reflection.

9. The answer here Infrared waves are Electromagnetic waves.

11. 

Net Displacment = 0

String becomes straight, No strain No. p.ε

12. Phase is different becuase all particles receive energy at different times.

13. $v = \sqrt{\dfrac{\text{Elastic property}}{\text{inertial property}}}$

$v = \left.\sqrt{\dfrac{\text{Elastic property}}{\text{inertial property}}}\right|$ not depends on vel. of source

15. The pulse does not have a definite wavelength or frequency, but has a definite speed of propagation (in non - dispersive medium).

16. Seismic wave is a mechanical wave.

17. X-Rays are electromangetic waves.

19. Sound waves in air are Longitudinal.

22. $y = 0.005 \sin(20\pi - 10\pi + \pi/2)$

$= 0.005 \sin(10\pi + \pi/2)$cm

$= 0.005 \sin(\pi/2)$

$y = 0.005$ cm

23. $V = \sqrt{\dfrac{\gamma RT}{M_\omega}} \Rightarrow V \alpha \dfrac{1}{\sqrt{M_w}}$

25. For oop $n_0 = \dfrac{V}{2\ell} = 550$ Hz

So, 1.1 KHz will be II Harmonic and COP and OOP of same length can't produce same frequency in any Harmonic case.

26. $n_A \sim n_B = 5$

by increasing the tension frequency will increase

$n_A \sim n'_B = 3$

If n_A = 427 Hz

So, that n_B = 422 Hz

30. Mechanical waves require inertia to propogate.

31. At a instant picture show position of med particle of that instant. It shows shape of wave.

32. For displacement to be same

$\sin(kx - \omega t) = \sin(kx^1 - \omega t)$

at t = 0 $\quad kx + 2n\pi = kx'$

$$x^1 = x + \frac{2n\pi}{k}$$

33. Longitudinal inside water and transverse on surface of water.

34. P-wave inside water & S-wave on surface of water.

35. Frequency of resultant wave = $n_{avg.} = \dfrac{n_1 + n_2}{2}$

36. $y_{in} = a\sin(kx - \omega t) = a\sin[-(\omega t - kx)]$

$y_{in} = -a\sin(\omega t - kx)$

$y_{ref} = -a\sin[\omega t + kx]$

$y_{sw} = y_i + y_r$

$= -a[\sin(\omega t - kx) + \sin(\omega t + kx)]$

$A_{sw} = 2a \cos kx$

37. When compression reaches open end it reflects as rarefraction.

38. In progressive wave Amplitude of all particles of med will be same.

39. In standing wave amp. of diff. particle is diff. max. at Antinode and min at node.

40. Wavelength changes only when source moves.

41. Pressure wave shows change in pressure.

P - wave and disp. wave has N_2 phase diff.

42. $\frac{\lambda}{4} = 25.5$ cm $\Rightarrow$ $\lambda = 102$ cm $= 1.02$ m

Now $v = n\lambda = (340)(1.02) \approx 347$ m/s

43. By, comparing $\lambda = 3$m, $\ell = \lambda/2$ one loop is formed therefore, all particle will oscillate in same phase.

44. $V \propto \frac{1}{\sqrt{P}}$

$P_{\text{moist air}} < P_{\text{dry air}}$

$V_{\text{moist air}} > V_{\text{dry air}}$

45. $V \propto \sqrt{T}$

$V = n\lambda$

$V \propto \lambda$

46. The time of compression and rerefaction is too small, i.e. we can assume adiabatic process and hence no transfer of heat.